CODILITY QUESTION

Task 1 English

A non-empty array A consisting of N numbers is given. The array is sorted in non-decreasing order. The *absolute distinct count* of this array is the number of distinct absolute values among the elements of the array.

For example, consider array A such that:

```
A[0] = -5
A[1] = -3
A[2] = -1
A[3] =  0
A[4] =  3
A[5] =  6
```

The absolute distinct count of this array is 5, because there are 5 distinct absolute values among the elements of this array, namely 0, 1, 3, 5 and 6.

Write a function:

```
int solution(vector<int> &A);
```

that, given a non-empty array A consisting of N numbers, returns absolute distinct count of array A.

For example, given array A such that:

```
A[0] = -5
A[1] = -3
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A[3] =  0
A[4] =  3
A[5] =  6
```

the function should return 5, as explained above.

Write an **efficient** algorithm for the following assumptions:

- N is an integer within the range [1..100,000];
- each element of array A is an integer within the range [−2,147,483,648..2,147,483,647];
- array A is sorted in non-decreasing order.

SOLUTION AND TEST INPUT

Files

- task1
  - solution.cpp
  - test-input.txt

solution.cpp | test-input.txt

```cpp
int solution(vector<int> &A) {
    int left = 0;
    int right = A.size() - 1;

    int count = 0;
    int last = -1;

    while (left <= right) {
        int leftVal = abs(A[left]);
        int rightVal = abs(A[right]);

        int current;

        if (leftVal > rightVal) {
            current = leftVal;
            left++;
        } else if (leftVal < rightVal) {
            current = rightVal;
            right--;
        } else {
            current = leftVal;
            left++;
            right--;
        }

        if (current != last) {
            count++;
            last = current;
        }
    }

    return count;
}
```

TERMINAL OUTPUT

Test Output

```
Compilation successful.

Example test:   [-5, -3, -1, 0, 3, 6]
OK

Your test case: [-5, -3, -1, 0, 3, 6]
Returned value: 5

Your code is syntactically correct and works properly on the example test.
Note that the example tests are not part of your score. On submission at least 8 test cases not shown here will assess your solution.
```

REPORT

Codility_

Screen Report: Anonymous
Test Name:

Check out Codility training tasks

Summary | Timeline

Tasks summary

Task	Effective time spent	Score
AbsDistinct C++	28 min	92%

Total score

92%

Assessment integrity No issues

See less

Integrity safeguards

Signal	Issues 0	Action
Network IP check The candidate used one network during the assessment	Not found	

Tasks Details

Tasks Details

Easy

1. AbsDistinct
Compute number of distinct absolute values of sorted array elements.

Task Score	Correctness	Performance
92%	90%	100%

Task description

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Write an **efficient** algorithm for the following assumptions:

- N is an integer within the range [1..100,000];

Solution

Programming language used:	C++
Time spent on task:	28 minutes
Notes:	*not defined yet*

Task timeline Preview

Code: 19:23:24 UTC, cpp, final, score: **92**

show code in pop-up

```
        int current;

        if (leftVal > rightVal) {
            current = leftVal;
            left++;
        } else if (leftVal < rightVal) {
            current = rightVal;
            right--;
        } else {
            current = leftVal;
            left++;
            right--;
        }

        if (current != last) {
            count++;
            last = current;
        }
    }

    return count;
```

18:55:59	19:23:24
Away from Codility tab	0%

Analysis summary

The following issues have been detected: wrong answers.

For example, for the input `[-2147483648, -1, 0, 1]` the solution returned a wrong answer (got 4 expected 3).

Analysis

Detected time complexity: **O(N) or O(N*log(N))**

expand all	Example tests	
▶ example example test		✓ OK
expand all	**Correctness tests**	
▶ one_element		✓ OK
▶ two_elements		✓ OK
▶ same_elements		✓ OK
▶ simple		✓ OK
▶ simple_no_zero		✓ OK
▶ simple_no_same		✓ OK
▶ simple_no_negative		✓ OK
▶ simple_no_positive		✓ OK
▶ arith_overlow		✗ WRONG ANSWER got 4 expected 3
▶ medium_chaotic1		✓ OK
▶ medium_chaotic2		✓ OK
expand all	**Performance tests**	
▶ long_sequence_no_negative		✓ OK
▶ long_sequence_no_positive		✓ OK
▶ long_sequence		✓ OK